

Software Reference Architectures - Exploring Their Usage and Design in Practice

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Abstract. Software reference architectures have been around for quite some years. They have been designed and used with varying success. We have conducted an exploratory survey among software architects and developers to establish the extent to which SRA have penetrated among practitioners and to identify the benefits and problems practitioners face when using and designing SRA. In this article, we present our findings.

Keywords: Reference architecture, software architecture, survey.

1 Introduction

Software reference architectures (SRA) have become one of the ways to address problems in managing complexity, size, and scope of architectures, and in managing the dynamics of the environment in which systems are designed [1]. While comprehensive methods have been defined for the design, evaluation, and application of concrete software architectures, SRA have received relatively less attention in literature [2], [3]. The reasons for this can be probably traced in an assumption that theory on software architectures is directly applicable to SRA.

In our practice, we have faced difficulties in working with SRA [4], [5]. Similar observations inspired several, recent, and with still preliminary results, research efforts [6], [7]. This triggered our interest in finding out what practitioners think of SRA. Do they design and use SRA? Do they benefit from them? Do they face problems in using them? Currently, literature does not provide insights into the status of SRA in the software community. Finding on the use of SRA are discussed in [8], but those are from a local and domain specific nature.

To answer our questions and establish an initial picture for the state of the practice of SRA, we have embarked on conducting an exploratory, cross-sectional, web-based survey on SRA, following the guidelines for data collection in software engineering disciplines discussed in [9-14]. In our survey, we targeted the following objectives:

1. *Building awareness in the community for the global status of SRA.*
2. *Building awareness for differences (or lack of such) in existing practices with SRA on the basis of geographical, experience, etc. specifics.*
3. *Building awareness for problems that practitioners experience and benefits they obtain in their work with SRA.*

In this paper, we present the findings from the survey. We explain our survey methodology in Section 2. In Sections 3, 4, 5, and 6 we present the results from the survey.

2 Survey Methodology, Setup, and Execution

2.1 Data Collection, Target Population, and Sampling

A survey provides a “snapshot of the situation to capture the current status” [12]. Furthermore, “explorative surveys are used as a pre-study to a more thorough investigation to assure that important issues are not foreseen ... the information is gathered and analyzed, and the results are used to improve the full investigation” [12]. This fully matched with our goals of investigating the trends and establishing the status of SRA. We have opted for a web-survey as a data collection method as it allows the collection of standardized data from a large, remotely-located population, which can be used for the establishment of the status and trends of SRA. An extensive discussion on the advantages and disadvantages of on-line surveys is provided in [13].

The target population was defined to be software architects and developers as these roles are the major stakeholder roles related to SRA. We aimed at respondents from two tiers of countries as defined in [15], i.e., Tier 1 representing major software exporting nations (e.g., USA, Germany, The Netherlands) and Tier 2 formed by emerging software exporting nations (e.g., Eastern European countries, developed Latin American countries). Our sampling approach was quota sampling (defined by our tier classification) augmented by convenience and snowball sampling [11].

2.2 Survey Design and Execution

The questions in our survey were designed on the bases of existing literature on SRA, e.g., [1, 4, 16]. We have grouped the questions in order to present to the respondents a clear survey structure. Group 1 questions collected information on the respondents’ background and context. Group 2 questions contained questions determining the interpretation of the respondents for the term “software reference architecture”. Group 3 questions investigated the experience of the respondents with usage of SRA, the benefits that they have obtained, and the problems that they have faced when using them. Group 4, respectively, investigated the design of SRA and the problems accompanying the design process. The latter two groups represented the core of the survey. Finally, we have added in Group 5 questions where the respondents had an opportunity to provide their opinions and thoughts on SRA. After addressing the feedback from four pilot respondents, we ended with 32 questions distributed in the five groups.

For the distribution of the survey, we have used contacts from our network in combination with posting the survey to professional forums. We have received in total 114 responses. The major set of reactions was received through personal contacts and from relatively small but active, professional LinkedIn groups. Thus, the response rate from the directly targeted population was high. The response rate from the indirectly targeted population was poor. Although a low response rate “affects the representativeness of the sample” [9], “if the objective is to understand trends, then low response rates may be fine. The homogeneity of the population, and the sampling technique used also affect the extent to which one can generalize the results of surveys” [9]. As we discuss in Section 3, the population was relatively homogeneous and our goal was indeed understanding trends, rather than establishing precise percentages.

3 Results from the Introductory Questions (Groups 1 and 2)

From the total number of responses, we have filtered out the responses that did not fit in our target population. This resulted in 90 valid responses¹. We received 61 responses from 11 countries from Tier 1 (Netherlands dominating the set of responses) and 29 responses from 5 countries from Tier 2 (dominated by Bulgaria and Chile) giving a reasonable response from each tier. The size of the companies of the respondents were equally distributed among large (>500 employees), medium (50-500), and small (<50). Respondents from Tier 1 were predominantly from larger organizations and respondents from Tier 2 from smaller organizations (which is a characteristic of the tiers). The major group of respondents were employed by a "Software and services" organization (59%), followed by "Financials" (13%), and other. The respondents were relatively experienced with architectures, with a mean value of 10 years of experience. On the basis of the answers of the respondents for their occupation, we have classified the respondents into: 70 architects and 20 developers.

We have used the definition provided by Wikipedia (alternative definitions are discussed in [16]) and asked whether the respondents agree with it: *"A reference software architecture is a software architecture where the structures and respective elements and relations provide templates for concrete architectures in a particular domain or in a family of software systems."* Out of the 90 respondents, 66 (73%) felt experienced with the term and agreed with the definition (of which 8 had minor remarks), 17 agreed with the definition, but indicated that to them the term was not too clear before reading the definition, 7 indicated that they were not aware of the term SRA. To improve the survey representativeness, the latter respondents were filtered out from the rest of the survey results. The high percentage of respondents experienced with the term indicates the deep penetration of the term in the architecting community. Notably, 5 of the 7 respondents unaware of the term SRA had less than 5 years of experience.

Six more respondents from the target population (but excluded from subsequent percentage calculations) agreed also with the definition. However, at a later, control

¹ Available at <https://sites.google.com/site/samuilangelov/ECSAvalid.xls>

question, these six respondents gave examples for SRA that did not fit in the definition, referring to frameworks, methods, modelling techniques (Zachman, TOGAF, ARIS, COBIT, ITIL). We concluded that although professionals tend to agree around a definition on SRA, the high abstraction of the definition, and the numerous related terms lead to differences in the interpretation of the term.

4 Results from the Questions on Usage of SRA (Group 3)

Propagation of SRA. Out of the 61 respondents who have answered the questions on the usage of SRA, 5 have indicated to have never used a SRA in their career (8%). All other respondents have used a SRA in their work at least 1-3 times (33%), 3-10 times (31%), and more than 10 times (28%). This leads us to conclude that SRA have become a popular tool used in the design of software architectures.

Scope and Origin of Used SRA. SRA with different scopes are used in practice with a preponderance of SRA defined specifically for the organization of the respondent (49%) or defined for a complete domain (43%). Intriguingly, professionals make use predominantly of SRA defined by their own organization (63%), followed by reference architectures defined by standardization (37%), IT-consultancy (27%), consortium (20%), software house (18%), or governmental organization (16%). The least used SRA are those defined from research organizations (4%). These results imply:

- Professionals have mostly used SRA defined for their own organization and by their own organization. However, some of these SRA were defined by the organization to be applied at a wider scope, e.g., the complete domain.
- Research organizations have a minimal contribution to the design of used SRA.

Goals When Using SRA and Benefits Obtained. Based on literature, we have defined a set of benefits introduced by reference architectures and asked the respondents to select those sought by them when using SRA and those that they have actually obtained from using SRA. In addition to our list of possible benefits (see Table 1), respondents have indicated also “education and training”, “risk reduction”, and “assessment tool”. Twelve respondents indicated that they are not currently using a SRA. Since we preferred recent and therefore more relevant experiences for this question, these respondents were cut from the sample, leaving 49 respondents.

A limited number (16%) of the respondents saw little or no benefits of the usage of reference architectures. For a number of aims (indicated as “n/a” in Table 1), we considered that achieving them is a direct consequence from the actual usage of the SRA. The question for the benefit achieved was therefore not asked in these cases. In all measured aims, the actual benefits scored substantially lower than those initially expected - the difference (reduction) is fairly stable at 20%. We have also surveyed the overall estimation on the improvement of the quality of the architectures and architecting processes when using a SRA - 43% of respondents saw improvements in the architecture quality and 37% in the architecting process. In only 29% of the cases, do SRA improve both the architecture quality and process – a rather low result given the fact that following best practices is leading when implementing SRA.

Table 1. Sought and actual benefits from the usage of SRA (percentages reflect the % of respondents who have mentioned a particular reason)

Goals when using SRA	Benefit sought	Benefit achieved
To follow best practices	78%	n/a
To speed up design work	61%	35%
To establish a common architecture strategy/vision	61%	41%
To ensure reusability	57%	37%
To ensure interoperability with other systems	55%	37%
To structure work (follow guidelines)	47%	n/a
To comply with a standard	45%	n/a
To improve communication in the organization	45%	22%
To decrease costs	31%	10%
To get inspiration	18%	n/a
To make use of the most novel design solutions	12%	n/a
For political reasons (externally imposed)	10%	n/a

Problems with Using SRA. Out of 56 responders to this question, 9 respondents (16%) did not experience any problems with the usage of reference architecture (all of them from Tier 1 countries, and 8 of them from a large organization) and 11 (from both Tiers) experienced only minor problems (20%). The major problems faced by the respondents were that SRA are often too abstract (34%), the lack of common interpretation of the term SRA (34%), bad documentation of SRA (23%), poor quality of SRA (16%), and being too specific/limiting (14%). With respect to the documentation, out of the 13 respondents who have indicated problems with it, 69% missed guidelines for the application of the SRA, 62% missed one or more viewpoints in the documentation, 54% missed the quality attributes aimed at by the SRA, 39% had difficulties with interpreting SRA, 23% saw the notation used as improper, 15% had problems with the extensive volume of the documentation of SRA and 7% with the incompleteness of the documentation. Last but not least, 7% (4 respondents) indicated other problems in free text. Remarkably, they had a common core: inability to apply the SRA due to the SRA complexity and/or reluctance by the stakeholders due to lack of clear benefits and therefore of motivation. While being a low percentage, this common response has been given in free text and therefore raises a question for its relevance to other respondents. This is further strengthened by the conclusions drawn in [8]. In conclusions, we infer that complexity, abstraction, documentation of SRA, and reluctance of stakeholders to consider SRA due to lack of clear benefits are problems when applying SRA. Also, the observation made earlier on the lack of common interpretation of the SRA concept in the domain receives a confirmation.

5 Results from the Questions on Design of SRA (Group 4)

Number of SRA Designed. Many practitioners declared that they have designed SRA - 35 out of the 53 respondents to this group of questions (32 of them were in an architect role). Most respondents have designed 1-2 architectures (30%) but also 3-5 architectures collected a substantial number of votes (23%).

Scope and Origin of SRA Designs. The respondents have designed SRA for their own organization (58%), a specific external organization (32%), a set of external organizations (29%), and a complete domain (16%).

Goals When Designing SRA. The stated goals for the design of reference architectures are manifold (see Table 2). The table shows a definite agreement on many of the goals, with 7 of them scoring more than 50%, and with ‘Providing best practices’ peaking at 71%. An important misalignment between designing SRA and using them is “To provide a high quality design solution for an innovative application”. In other words, although many designers aim at it, innovation is not really appreciated by stakeholders embarking on the application of SRA. “Improving communications” and “decreasing costs” stand out as even more distant if we compare aims of designers with the actual benefits obtained from using SRA.

Table 2. Goals when designing SRA versus goals when using SRA (based on 31 respondents that have recently used and designed a SRA)

Goals when designing SRA	When designing SRA	When using SRA
To provide best practices	71%	77%
To promote reusability	58%	55%
To promote interoperability	58%	55%
To set up a common architecture strategy/vision	58%	65%
To improve communications	58%	55%
To set up a standard	52%	55%
To structure the work of designers	52%	45%
To speed up design work	48%	61%
To provide a high quality design for an innovative application	42%	7%
To decrease costs	36%	23%
To provide an inspiration tool to designers	13%	16%
For political reasons (externally imposed)	13%	10%

Usage of Designs. About 40% of the respondents could not estimate the usage of the SRA they have designed and 13% were not aware of any usage of some of the SRA they have designed. This disconnection prevents designers from getting feedback on their designs and on maintaining and improving the architectures.

Problems with Design SRA. The answers of the 35 respondents to this question depict a multitude of problems when designing SRA (Figure 1). Problems with stakeholders’ involvement led in the responses. They are followed by the lack of suitable

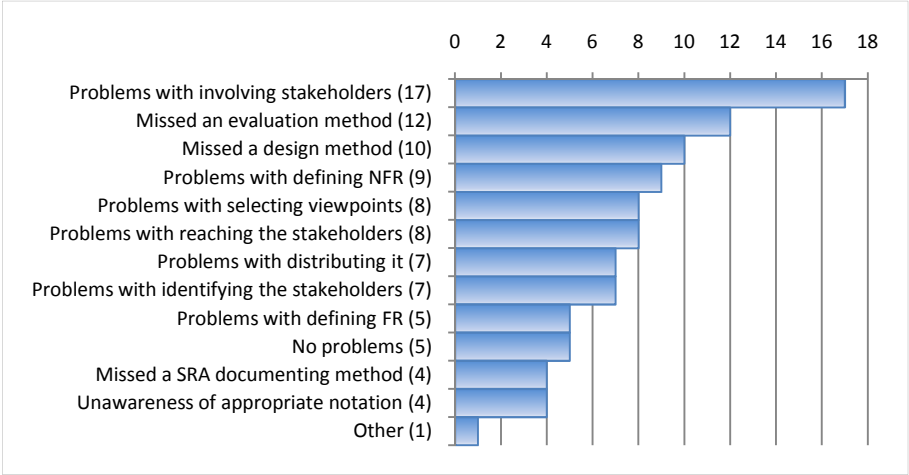


Fig. 1. Problems encountered during the design of SRA

methods for the design and evaluation of reference architectures. Only 14% did not report on experiencing problems. Figure 1 leads to the conclusion that professionals face multiple challenges during the design of SRA and existing theory and practices are not sufficient to facilitate their work.

6 Results from Questions on Status and Future (Group 5)

Predominantly, the respondents think that more and more SRA are being designed (51% of the respondents to this question, where 17% expressed the counter opinion). Similarly, 55% of the respondents in the last group of questions expect that the role of SRA will become more important in the future and only 13% expect that it will become less important. The majority of the respondents indicated that they are interested in the outcomes of the survey and that they are ready to be approached for clarification questions. From this, we conclude that there is an interest in the community in SRA and practitioners believe that SRA may even further grow in their importance.

7 Concluding Remarks

The survey improves the awareness in the community on state of the practice of SRA and the strengths and weaknesses of SRA. It indicates that SRA are widely recognized and used by practitioners but that there is still a lack of clear and commonly accepted definition of SRA which undermines progress in the field. The results delineate the leading benefits sought and obtained when using SRA as well as the problems encountered during the design and usage SRA. Some of the problems observed in the results are the misalignment between SRA design goals, usage goals, and eventual benefits obtained and the underperformance of SRA in terms of architecture quality and documentation. Stakeholder involvement and lack of design and evaluation

methods lead in the reported SRA design problems. No statistically valid specifics of the clusters in our population could be derived based on the number of responses.

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